

Implementing 2x2 CGE model with GEMPACK

file

MDATA # data file #;
set IND (i1,i2);

variable

(all,i,IND) x(i) # output of domestic industry #;
(all,i,IND) l(i) # labor in domestic industry #;
(all,i,IND) k(i) # capital in industry #;
(all,i,IND) p(i) # price of commodity by industry #;
(all,i,IND) q(i) # household demand for commodity #;
w # nominal wage #;
r # rent of capital in industry #;
y # household total income #;
ls # household supply of labor #;
ks # household supply of capital #;
(all,i,IND) c(i) # unit cost #;

coefficient

(all,i,IND) SL(i) # cost share of labor in industry #;
(all,i,IND) VX(i) # value of production of industry #;
(all,i,IND) VL(i) # labor cost of industry #;
(all,i,IND) VK(i) # industry capital #;
VLS # value of labor supply #;
VKS # value of capital supply #;
VY # value of total income #;
(all,i,IND) SIGMA(i);

read VL from file MDATA header "VL";
read VK from file MDATA header "VK";

update

(all,i,IND) VL(i) = w*l(i);
(all,i,IND) VK(i) = r*k(i);

formula

(all,i,IND) VX(i) = VL(i) + VK(i);
(all,i,IND) SL(i) = VL(i)/VX(i);
VLS = sum{i,IND,VL(i)};
VKS = sum{i,IND,VK(i)};
VY = VLS + VKS;
(all,i,IND) SIGMA(i) = 1;

equation

e_l # demand for labor #
(all,i,IND) l(i) = x(i) - SIGMA(i)*[w - c(i)];
e_k # demand for capital #
(all,i,IND) k(i) = x(i) - SIGMA(i)*[r - c(i)];
e_wave # price of primary factor composite #
(all,i,IND) c(i) = SL(i)*w + (1 - SL(i))*r;
e_pf # zero profit in production #
(all,i,IND) VX(i)*[p(i) + x(i)] =
VL(i)*[w + l(i)] + VK(i)*[r + k(i)];
e_q # demand for commodities #
(all,i,IND) q(i) = y - p(i);
e_y # household factor income #
VY*y = VLS*(w + ls) + VKS*(r + ks);
e_mc # market clearing for commodities #
(all,i,IND) x(i) = q(i);
e_ls # labor market clearing #
VLS*ls = sum{i,IND,VL(i)*l(i)};

Exercise on implementing 2x2 CGE model

1 Creating model file

Prepare and name a new empty folder/directory in your hard drive. Open TABMATE, create a new tablo file [File/New Tab File]. Save the file first as **model2x2.tab**, then write the code on the left column. When writing this code, make sure you understand their meaning. Especially in the equation part you can refer to the original equation written in standard mathematical form (Note that one of the market clearing condition equation, in this case, capital market clearing, is removed by Walras Law).

Tips: While writing the code, press F4 or Tablo Check button to check whether you made any mistake (syntact error).

When you finish, save again and press F11 or click the button **Tablo Code**.

By pressing F11/Tablo Code you are “implementing the tablo file”. Gempack will create three files needed for later simulation purpose. You have to do this again anytime you did modification to the tablo file.

- ▶ $l_i = x_i - \sigma_i(w - c_i)$
- ▶ $k_i = x_i - \sigma_i(r - c_i)$
- ▶ $c_i = S_i^L w + (1 - S_i^L) r$
- ▶ $P_i X_i(p_i + x_i) = W L_i(w + l_i) + R K_i(r + k_i)$
- ▶ $q_i = y - p_i$
- ▶ $Y y = W L^S \cdot (w + l^S) + R K^S \cdot (r + k^S)$
- ▶ $x_i = q_i$
- ▶ $W L^S \cdot l^S = \sum_i W L_i \cdot l_i$

2. Creating model's database

Database file record the base coefficient for the model. In this case, there are two base coefficient i.e., $W L_i$ and $R K_i$. In Gempack the database file has the extension .har and created and read by Gempack program called VIEWHAR.

1. Open VIEWHAR program.
2. Create a new file (File/Create New File). If Gempack is just recently installed, the Create New File menu will not appear. In this case, you should first click Use Advanced Editing Menu in the File menu.
3. Create two header for the two base coefficient (Edit/Create New Header). Note: For $W L_i$ the header type is Real [RE], the 4-character header is VL, Default value is 0.0, Longname “Labor cost of Industry”, No of dimension is 1, and size is 2. For $R K_i$, the header is VK and the longname is “Capital cost of Industry”. The rest is the same.
4. Open file **io.xls**, and by copy and paste enter the value of $W L_i$ and $R K_i$ from the excel file to the har file.
5. Save the database file as model2x2.har

3. Creating simulation file

Simulation file is a text file with extension .CMF. In this file we will have instruction of how to shock the model (shock statement) also the closure of the model. Model's closure is a statement indicating which variables are endogenous and which variable are exogenous to make sure that the number of endogenous variables is equal to the number of equations. Note: We can only shock exogenous variables.

1. Open TABMATE program.
2. Create a new CMF file (File/New CMF File).
3. Type the code below:

```
Auxiliary file = model2x2;  
File MDATA = model2x2.har;  
Updated File MDATA = model2x2.upd;  
Method = Gragg;  
Steps = 2 4 6;  
Solution file = <cmf>.sl4 ;  
Verbal Description = Simulation with 2x2 CGE model;  
exogenous ks ls p1("i1");  
Rest endogenous;  
Shock p1("i1") = 1.23;
```

4. Save the file as sim2x2.cmf
5. In the simulation file, we list variable k^s , l^s , and p_1 . The rest of the variables then are treated as endogenous variables. We choose p_1 or $p1("i1")$ as the model's numeraire, and we want to check the model's **nominal homogeneity** by shocking p_1 by 1.23%. It should be expected that all nominal variables in the model will change by 1.23% while all of the real/quantity variable will not change at all.
6. You can also test for **real homogeneity test** by shocking all exogenous quantity variables by the same amount of shock. The model will pass this test if all the quantity/real variables in the model change by the same amount of shock and all the price variables will not change at all. **Try to do this!**

4. Policy simulations

The impact of sea level rise

The island is facing climate change threat. In the near future, sea level rise will reduce the land area by half (50 percent). With the help of production possibility frontier curves, explain the results (for example, why one industry expand more than the others, or why price of one commodity increase/decrease more than the others).

			Sector	
			i1	i1
Output	x	x_i		
Price	p	p_i		
Capital	k	k_i		
Labor	l	l_i		
Capital Supply	k^s	k^s		
Labor Supply	l^s	l^s		
Wage	w	w		
Rent	r	r		
Income	y	y		

You can explain and illustrate your answer here: (Note that your answer to x_1 and x_2 should be identical with the one explained in the previous lecture!)

5. Modification of the model

Add two variables to the model (1) real GDP; (2) consumer's price index (CPI) to the model, and run again your policy simulation exercise and look at what happen to these two new variables. Does it make sense?